

Relationship Table

Movie Ticket Booking System

Relationship Analysis

The relationships are selected using the lifetime test specified in the assignment:
if the whole object is destroyed, does the part also cease to exist?

Pair	Relationship	Justification
Cinema — Screen	Composition	Cinema owns its Screen objects. If the Cinema is destroyed, its Screens are also destroyed; therefore their lifetimes are dependent.
Screen — Seat	Composition	Screen owns its physical Seat objects. If the Screen is destroyed, its physical Seats are no longer part of that auditorium.
Show — Movie	Aggregation	Show uses an existing Movie, but the Movie can exist independently of a particular Show. Destroying a Show does not destroy the Movie.
Show — Screen	Association	A Show is scheduled on an existing Screen. The Screen can host other Shows and has an independent lifetime.
Show — ShowSeat	Composition	ShowSeat represents seat status specifically for one Show. Its lifetime is dependent on that Show.
Booking — Customer	Association	A Customer exists independently of a particular Booking and can make multiple bookings. Destroying a Booking does not destroy the Customer.

Booking — ShowSeat	Aggregation	Booking refers to existing ShowSeat objects. ShowSeat belongs to the Show and continues to exist independently of a Booking, including after cancellation.
Booking — Payment	Association	Booking uses Payment to complete the transaction, but the payment object is not owned by the Booking in terms of lifetime.
Payment — UpiPayment	Inheritance	UpiPayment is a specialized Payment and implements the abstract payment contract.
Payment — CardPayment	Inheritance	CardPayment is a specialized Payment and implements the abstract payment contract.
Payment — CashPayment	Inheritance	CashPayment is a specialized Payment and implements the abstract payment contract.
BookingService — Booking	Association	BookingService orchestrates the booking process and works with Booking objects; it does not own their lifetime.

Relationship Summary

Relationship Type	Relationships
Composition	Cinema → Screen; Screen → Seat; Show → ShowSeat
Aggregation	Show → Movie; Booking → ShowSeat
Association	Show → Screen; Booking → Customer; Booking → Payment; BookingService → Booking
Inheritance	Payment → UpiPayment; Payment → CardPayment; Payment → CashPayment

Lifetime Test

Composition: the part's lifetime depends on the whole.

Aggregation: the part can exist independently of the whole.

Association: objects interact or refer to each other without ownership of lifetime.

Inheritance: the child is a specialized form of the parent and follows its contract.